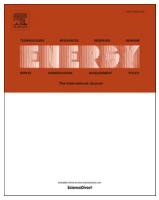
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Household, sociodemographic, building and land cover factors affecting residential summer electricity consumption: A systematic statistical study in Phoenix, AZ

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ABSTRACT

Understanding determinants of residential electricity consumption is crucial for urban sustainability efforts for planners and policy makers to develop targeted strategies to lower energy use, reduce greenhouse gas emissions, and to increase community resilience. This study presents a systematic approach to build an interpretable multivariate linear model, addressing challenges like outlier detection, multicollinearity, non-normality, and heteroscedasticity. Using 2019 summer residential electricity data for 426 census tracts in Phoenix and 30 variables, the approach involves (1) addressing multicollinearity and regression outliers through Variance Inflation Factor and studentized residual analysis, (2) comparing an automatic variable selection method with Ridge, Lasso, and Elastic Net regression, (3) evaluating the final model, and (4) interpreting variable effects. Critical findings reveal multicollinearity in land cover and racial variables, while 21 census tracts on the urban periphery exhibit outliers with unique features. Variable selection demonstrates the significance of household and building information in influencing residential electricity consumption. Household variables alone account for 84 % of electricity usage variation. Incorporating building information and land cover variables reduces errors by 35 % and 26 % respectively, emphasizing the significance of including household characteristics as predictors or control variables when modeling electricity consumption. A final model with 93 % explanatory power enables precise predictions.

1. Introduction

Buildings are widely recognized as significant contributors to carbon emissions because of the use of energy to heat, cool, and operate appliances [1]. In the context of sustainable urban development, residential energy consumption therefore becomes a key focus, highlighting the imperative to understand its determinants for effective energy-efficiency strategies in urban areas. Recognizing that we lack a single-strategy approach, solutions require a comprehensive understanding of how socio-demographic and economic factors interact with energy consumption [2], underscoring the importance and necessity of robust statistical modeling in urban sustainability endeavors.

The literature on the determinants of electricity use collectively emphasizes the intricate nature of electricity consumption patterns influenced by building characteristics, socio-demographics, and urban contexts. Data-driven models for residential energy consumption extensively incorporate census household and demographic information to unveil consumption patterns. In the realm of statistical models for residential electricity use, the importance of socio-demographic variables is emphasized in various studies [3]. Recent city-focused investigations, like that of Kontokosta et al. [4] for New York City, underscore a direct correlation between income and household electricity consumption, particularly in low-income households, which tend to consume this form of energy inefficiently. The study from Kim et al. [5] compares linear and nonlinear models using multiple linear regression and decision trees, identifying that building systems type, and occupant characteristics significantly affect electricity usage. Lawal's et al. [6] research in Atlanta employs orthogonalization algorithms, machine learning, and variable selection to predict electricity use, emphasizing the importance of socio-demographic factors. Hsu [7] and Satre-Meloy [8] use regularization methods in New York City and Palo Alto, California, respectively, highlighting variables such as occupancy,

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household size, and income as electricity consumption predictors. Xu's et al. [9] analysis of a large electricity consumption data set in residential buildings in the U.S, reveals the considerable impact of age, household size, income, education level, and occupancy type.

While these studies provide valuable insights into socio-demographic aspects of electricity use, challenges arise when incorporating building information data into data-driven models. Describing elements like wall insulation, glazing, heating, ventilation, and air conditioning systems (HVAC), and appliances presents significant challenges. Liu et al. [10] investigate into the impact of key drivers on nonresidential buildings' electricity consumption in Chongqing, China, using a data-driven approach integrating regularization and quantile regression methods. The study pinpoints building characteristics, including gross floor area, number of occupants, and cooling type, as key drivers with heterogeneous effects on electricity usage across different quantiles. Similarly, McKenna et al. [11] investigates daily electricity and gas demand in British housing, leveraging linked smart meter and socio-technical data. Employing linear mixed effects modeling, the study identifies factors as building age, number of rooms and air conditions ownership as highly associated with variations in daily total electricity consumption. Despite challenges, a consistent theme in research indicates that building characteristics [12] and socio-demographic factors [13] are pivotal predictors in modeling household electricity consumption. Methodologically, studies utilize various statistical techniques, including regression analysis and statistical learning methods, to model energy use (electricity and gas), identify significant predictors, and explore interactions among variables. Limitations include modeling energy use for specific periods and challenges in collecting high-temporal-resolution metered energy use data due to privacy concerns.

Notably, variables describing household characteristics other than socio-demographics have received less attention in city-level energy consumption studies. In regional scales, O'Neill and Chen [14] finding that householder age and number of children per household impacts energy use (by fuel types) across the U.S. underscores the importance of exploring such variables. Similarly, Brounen et al. [2] analyzed residential electricity and gas use in the Netherlands, emphasizing the economic behavior of households. The study suggests that residential gas consumption is influenced by structural dwelling characteristics, while electricity consumption correlates more directly with household composition, income, and family structure. Lenzen et al. [15] takes a global perspective, using per capita energy requirements (accounting for all sectors) to quantify environmental impacts. The findings underscore the varied effects of increasing income on household energy requirements across different countries. In general, those consistent findings across studies makes clear the importance of household occupancy, size, and income in explaining residential energy consumption, particularly electricity [16]. Research by Abrahamse and Steg [13] delves into the usefulness of sociodemographic data at the household level, showcasing its role in, not only modeling electricity and gas use, but also exploring occupants' intentions to reduce consumption by incorporating psychological variables. Additionally, contextual factors such as land cover characteristics appear to be overlooked.

Delving into building urban energy modeling (UBEM), the literature outlines two primary approaches: Top-Down and Bottom-Up models [17]. Top-Down models rely on aggregated historical energy data for cities or regions, employing statistical models (e.g. Ref. [18]). Conversely, Bottom-Up models incorporate detailed spatiotemporal data, leveraging statistical methods and machine learning to complement physics-based building energy models [19]. Sovacool et al. [20] advocates for interdisciplinary, problem-oriented research to address the challenges of building a low-carbon future energy systems, highlighting the potential integration of social science quantitative methods into energy modeling frameworks. Studies like those made by Hong et al. [21], Ferrando et al. [22] and Ang et al. [23] underscore the value of incorporating sociodemographic insights into UBEM, enhancing predictions, and understanding factors influencing energy consumption in urban areas. The literature underscores the importance of accurately estimating the effects of socio-demographics, particularly in variables describing household characteristics, and considering the potential impact of other contextual factors like land cover descriptors on building electrical energy consumption.

The objective of this study is to present a systematic and comprehensive statistical approach, for the purpose of selecting optimal subsets of predictors, thereby creating an interpretable multivariate linear model. This approach leverages an automatic variable selection technique and regularization methods, aiming to address prevalent challenges in linear statistical models. These challenges include outlier detection, overfitting, multicollinearity, non-normality, and heteroscedasticity in the residuals. This research contributes methodologically by providing a clear exposition of statistical modeling techniques employed to diagnose and tackle these issues. The focus is on constructing interpretable and predictive linear models, particularly when dealing with aggregated census tract information, with a specific emphasis on modeling residential electricity usage. Furthermore, the study enriches the variable selection and regularization literature within fields such as residential building stock energy modeling and urban building energy modeling following data driven approaches. On an empirical level, the paper contributes by utilizing a diverse set of over 20 predictors categorized into household, building, land cover, and racial parameters. These predictors are employed to model metered residential electricity consumption obtained from a utility company. Through this empirical investigation, the study aims to offer insights into the intricate relationships between various parameters and electricity usage, adding valuable knowledge to the broader discourse on residential energy modeling.

2. Methods and modelling framework

To establish a conceptual foundation for the statistical approaches employed in this study and to elucidate their interconnections, we present an overview in the following sections. We begin with a concise exposition of how we address multiple linear regression limitations in terms focusing on multicollinearity and influential observations. Next, we research procedures for subset selection and regularization techniques, elucidating their relevance to our methodology. Concluding the methodology, we outline the validation procedure and metrics applied in this study.

2.1. Multicollinearity, outliers, and interpretability in ordinary least squares regression

Expressed in matrix notation, the first-order additive multiple regression model is defined as:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \tag{1}$$

where the dependent variable $\mathbf{Y}$ is an n x 1 vector that contains Y_i for the i-th observation, the k independent variables (X_{ik}) are contained in $\mathbf{X}$ as an n x p matrix, $\boldsymbol{\beta}$ is a p x 1 vector with the coefficient parameters; β_0 is the intercept and β_k is the slope for the *k-th* predictor. Assumptions on the random error term $\boldsymbol{\varepsilon}$ (n x 1 vector) include independent and identically distributed, with normal distribution, mean zero, and constant variance σ^2 (ε *iid* $\sim N(0, \sigma^2)$) [24]. Variation in Y_i is measured by the deviations of the independent variable observations around its mean ($\overline{Y}$). The sum of the square of this total deviation is the total sum of squares ($SSTO = \sum (Y_i - \overline{Y})^2$), which is partitioned into regression sum of squares ($SSR = \sum (\widehat{Y}_i - \overline{Y})^2$) and error sum of squares ($SSE = \sum (Y_i - \widehat{Y}_i)^2$). The least square estimator used for parameter estimation ($\widehat{\boldsymbol{\beta}}$) to minimize SSE is:

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\,\mathbf{X}'\mathbf{Y} \tag{2}$$

The multiple standardized regression model applied correlation transformations [25] to the variables to aid comparison and interpretation [26]. Using the transformed variables in equation (2) to estimate the standardized parameters ($\widehat{\boldsymbol{\beta}^*}$), make that $\mathbf{X}'\mathbf{X}$ becomes the correlation matrix of independent variables (r_{xx}), and $\mathbf{X}'\mathbf{Y}$ now contains the correlation coefficients between $\mathbf{Y}$ and the predictors (r_{yx}). These are essential for posterior analyses. Note that $\widehat{\boldsymbol{\beta}^*}$ are used for predictor effects comparisons. However, this practice has faced criticism from certain authors such as Greenland et al. [27] and Fox et al. [28], due to its impact on β_0, which is drop from the model, and the standard errors of transformed parameters, potentially distorting coefficients. This issue is particularly pronounced when dealing with round-off errors in the presence of moderately or highly linearly correlated predictor variables, an issue known as multicollinearity, a common challenge in regression analysis. Multicollinearity can be quantified by obtaining the *k-th* element of the diagonal of the correlation matrix inverse ($\mathbf{r}_{XX}^{-1}$) which contains the Variance Inflation Factors (VIF). The VIF for the *k-th* regressor (VIF_k) can be computed as [29]:

$$VIF_k = \frac{\left[(\boldsymbol{X}'\boldsymbol{X})^{-1}\right]_{kk}}{\mathbf{r}_{XX}^{-1}} \tag{3}$$

The *k-th* entry on the main diagonal of $(\mathbf{X}'\mathbf{X})^{-1}$ is denoted as $\left[(\boldsymbol{X}'\boldsymbol{X})^{-1}\right]_{kk}$. Correlations among variables inflate the predictor's variance, resulting in VIF values exceeding 1. As a practical guideline, a VIF greater than 10 signals multicollinearity [30]. In high-dimensional datasets, identifying outliers individually for each variable can lead to significant data loss. Instead, we focus on detecting influential points in multivariate regression, which can impact least squares estimators. The hat matrix (H) [31] is vital in this analysis:

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\,\mathbf{X}' \tag{4}$$

Considering that fitted values ($\widehat{\mathbf{Y}}$) are obtained as the product of $\mathbf{H}\,\mathbf{Y}$, a raw residuals vector ($\mathbf{e}$) is directly computed. To measure an observation's influence on residuals [32], we compute studentized deleted residuals (t_i) [33]:

$$t_i = e_i\sqrt{\frac{n-p-1}{\mathrm{SSE}(1-h_{ii})-e_i^2}} \tag{5}$$

where e_i is the ith raw residual in $\mathbf{e}$, and h_{ii} is the diagonal element of $\mathbf{H}$. Visual inspection of larger t_i values help identify potential outliers with respect to Y. Leverage points, identified using h_{ii}, indicate how far predictor variables values deviate from their mean. Moderate leverage typically falls between 0.2 and 0.5 [24]. Observations can be outliers with respect to both Y and X, impacting predictor coefficients and fitted values. For this, Cook's Distances (D_i) provide a metric for detecting outliers and influential observations [34]:

$$D_i = \frac{e_i^2(n-p)}{p(\mathrm{SSE})}\left(\frac{h_{ii}}{(1-h_{ii})^2}\right) \tag{6}$$

Ensuring that the final regression models and their variables are interpretable is crucial for this study. Centering variables aids interpretation by reducing multicollinearity [35]. For a single predictor variable X_k, the centered predictor variable ($\widetilde{X}_{ik}$) at the *i*-th observation is:

$$\widetilde{X}_{ik} = X_{ik} - \overline{X}_k \tag{7}$$

Interpretation becomes more straightforward because a one-unit change corresponds to a one-standard-deviation change in the original variable, with the intercept (β_0) representing the expected response variable value when all predictors are at their mean. Coefficients (β_k) now represent the change in the response variable for a one-unit change in the centered predictor, holding other predictors constant.

2.2. Best subset and regularization

In statistical modeling, finding a concise model that explains variation in the dependent variable with minimal predictors is crucial. However, dealing with high-dimensional datasets poses challenges due to the impracticality of fitting every possible regression model. To address this, the best subset regression, based on the leaps and bounds algorithm [36], is used. It employs the branch-and-bound technique to ensure exhaustive searches of all possible combinations [37]. This automatic variable selection technique is validated using performance metrics based on SSE to select the final model.

On the other hand, regularization methods aim to reduce the variance of regression coefficient estimates by incorporating penalty terms that "shrink" coefficients toward zero. This shrinkage not only diminishes variance but also introduces bias in regression coefficients, illustrating the bias–variance tradeoff. It is essential to only apply shrinkage to predictor coefficients, not the intercept, making the use of the standardized regression model ideal [38].

The ridge estimator, introduced by Hoerl and Kennard [39], modifies the least square estimators using the standardized variables:

$$\widehat{\boldsymbol{\beta}^R} = (\mathbf{r}_{xx} + \lambda\mathbf{I})^{-1}\,\mathbf{r}_{yx} \tag{8}$$

here, $\widehat{\boldsymbol{\beta}^R}$ represents the vector of standardized ridge regression coefficients, $\mathbf{I}$ is the $(p-1)$ x $(p-1)$ identity squared matrix and λ is the biasing constant or regularization parameter >0. λ introduces bias, reducing variance and stabilizing coefficients. Ridge regression is advantageous in addressing correlated predictors, as it decreases VIF values [40].

The optimization problem for ridge regression is to minimize:

$$\sum_{i=1}^{n}(Y_i - \widehat{Y}_i)^2 + \lambda\sum_{k=1}^{p}\beta_k^2 = SSE + \lambda\sum_{k=1}^{p}\beta_k^2 \tag{9}$$

where the term $\lambda\sum_{j=1}^{p}\beta_k^2$ is an l_2 penalty that consist in the squared sum of coefficients, this makes that the ridge penalty never shrinks a coefficient to zero. In contrast, Lasso regression, using an l_1 penalty (the absolute sum of the coefficients), aims to force some predictor coefficients to zero [41]. In lasso regression, the optimization problem consists in finding the lasso regression coefficients ($\widehat{\boldsymbol{\beta}^L}$) that minimize:

$$SSE + \lambda\sum_{k=1}^{p}|\beta_k| \tag{10}$$

This makes lasso regression an appropriate alternative when some predictors are expected to be irrelevant since an automatic variable selection is performed by dropping out predictors which its lasso coefficient is zero, promoting sparsity [42].

For situations with many correlated predictors and a significantly large number of predictors, Elastic Net regression combine l_1 and l_2 penalty terms [43]. The minimization problem that elastic net regression coefficients ($\widehat{\boldsymbol{\beta}^E}$) should satisfy is:

$$SSE + \lambda_1\sum_{k=1}^{p}|\beta_k| + \lambda_2\sum_{k=1}^{p}\beta_k^2 \tag{11}$$

Both λ_1 and λ_2 control the respective penalties, providing a flexible approach. The regularization parameter λ must be tuned to obtain an accurate regression model. Cross-validation is crucial for selecting the final model, with different λ values producing distinct coefficient vectors ($\widehat{\boldsymbol{\beta}^R}$, $\widehat{\boldsymbol{\beta}^L}$ and $\widehat{\boldsymbol{\beta}^E}$).

2.3. Cross-validation and model selection

The mean square error (MSE) is a critical measure for assessing coefficient estimates' quality [44], balancing variance, and bias in regression models [38]. It evaluates model performance by comparing predicted and actual values, calculated as the SSE divided by degrees of freedom ($n - p$). Minimizing MSE is the primary criterion for model selection. To address dataset limitations, such as small sample sizes, Leave-One-Out Cross-Validation (LOOCV) is utilized [45]. LOOCV iteratively validates the model by using one observation for validation while training on the remaining data, providing estimates for both in-sample and out-of-sample MSE [42]. This technique prevents overfitting, ensuring models fit the true relationship rather than noise, by comparing in-sample and out-of-sample MSE. Effective models perform well on both training and validation metrics.

To compare selected models' goodness of fit, the root mean square error (RMSE) as a measure of prediction accuracy is used. RMSE provides insight into the typical magnitude of errors in predictions and penalizes larger errors more heavily. The coefficient of determination (R^2) and its adjusted version are also employed. R^2 expresses the proportion of Y variation explained by the model, while adjusted R^2 addresses the issue of favoring high-dimensional models [24]. It balances model accuracy with the number of predictors. R^2 and adjusted R^2 are estimated as fallow:

$$R^2 = 1 - \frac{SSE}{SSTO} \tag{12}$$

$$Adjusted\ R^2 = 1 - \frac{\frac{SSE}{n-p}}{\frac{SSTO}{n-1}} = 1 - \frac{MSE}{MSTO} \tag{13}$$

where MSTO is the mean sum of squares. A metric that measures the contribution of specific predictors is the coefficient of partial determination ($R^2_{YA|B}$). It quantifies the proportion of unexplained Y variation reduced when introducing a group of variables to an existing regression model [46]. $R^2_{YA|B}$ is done by splitting the k predictors into an A (u) and B (k - u) groups. The marginal contribution of variation that is explained by the variables in group B that is not explained by the predictors in group A is measure by:

$$R^2_{YA|B} = \frac{SSE(A) - SSE(A,B)}{SSE(A)} \tag{14}$$

The partial $R^2_{YA|B}$ metric becomes crucial when selecting the final multivariate regression model generated from variable selection and regularization procedures. It facilitates the quantification of the relative contribution of a specific type of predictor. This is particularly important as some variables selected through automatic procedures might not substantially reduce variability in Y.

2.4. Modeling framework and software utilization

The modeling approach in this study follows a systematic four-step process to enhance the interpretability and robustness of the multivariate linear model predicting residential electricity consumption. In the first step, we carefully handle multicollinearity by analyzing Pearson correlation coefficients and Variance Inflation Factors (VIF). We identify significant correlations and strategically remove highly correlated predictors to mitigate multicollinearity issues and ensure analysis stability. This results in a subset of independent variables with VIF<10. Additionally, we meticulously examine regression outliers, leverage, and influential observations to address potential sources of non-normality and non-heteroscedasticity in the residuals. In step 2, we use variable selection and regularization techniques to identify the most relevant predictors. The LOOCV method helps determine optimal regularization parameters for Ridge, Lasso, and Elastic Net methods, using Mean Squared Error (CVMSE) as a metric to identify the best models. We compare these models using both in-sample and out-of-sample goodness-of-fit metrics. These findings inform the selection of a final model, striking a balance between model complexity and predictive performance.

In the third step, we thoroughly evaluate the goodness of fit and performance of the final model. We construct different models based on predictor categories, providing insights into how much these contribute to the predictive power of the final model. By examining adjusted R^2 values and partial R^2, we gain a relative understanding of each variable's categories impact on explaining variations in residential electricity consumption. The final step focuses on interpreting parameter coefficients in the selected final model, highlighting variables significantly influencing electricity consumption variations. Examination of variable contributions ensures a comprehensive understanding of the model's performance in capturing household sociodemographic, building and land cover factors affecting electricity consumption.

This study used the statistical computing software R [47] for data processing and analysis. Key packages included the *car* package [28] for calculating Variance Inflation Factor (VIF) values and diagnostic measures, the *caret* library [48] for LOOCV, the *glmnet* package [49] for regularization techniques, the *leaps* package [50] for Best Subset and Stepwise Backward selection procedures, the *stargazer* package [51] for comparing regression coefficients, and the *rsq* package [52] for calculating the coefficient of partial determination ($R^2_{YA|B}$).

3. Study area and data

In the following sections, we present an overview of the Phoenix Metropolitan Area, outlining its key urban and climatic features. Additionally, we provide a detailed description of the dependent variable, average summer residential electricity use, and explain the categories of predictors analyzed in this study. Descriptive statistics for each variable are included to offer a clear understanding of the dataset.

3.1. Residential, household and climate profiles of the Phoenix metropolitan area

The Phoenix metropolitan area, or PMA (Fig. 1a), encompasses Phoenix, the capital and largest city in Arizona, along with more than 20 adjacent cities, including Mesa, Chandler, Scottsdale, Tempe, and Glendale. Experiencing substantial population growth and urban development in recent decades, the housing landscape features a mix of single-family homes, apartment complexes, and gated communities, contributing to suburban sprawl [53]. The residential profile indicates a diverse mix of families, young professionals, and retirees, with a notable preference for homeownership [54] and a desire for single-family homes [55]. The climate in the PMA is characterized by hot desert conditions. Summers are intensely hot, with daytime temperatures often exceeding 100 °F (37.8 °C), accompanied by low humidity and clear skies. Winters are mild, creating a distinctive climate profile for the region [56]. According to Jones et al. [57], in the next 50 years vulnerable populations, particularly the elderly and low-income households, might face disproportionate increases in electricity cost, and racial minorities will continue to experience significant affordability issues. Equally important, Wentz et al. [58] explore community resilience in Phoenix, focusing on rising urban temperatures and the COVID-19 pandemic. Analyzing factors affecting residential electricity use, the authors found home structure significantly influences consumption, especially evident in lower-income areas. While work-from-home policies did not notably impact usage, household size became more influential in 2020, highlighting the importance of recognizing home structure in electricity consumption.

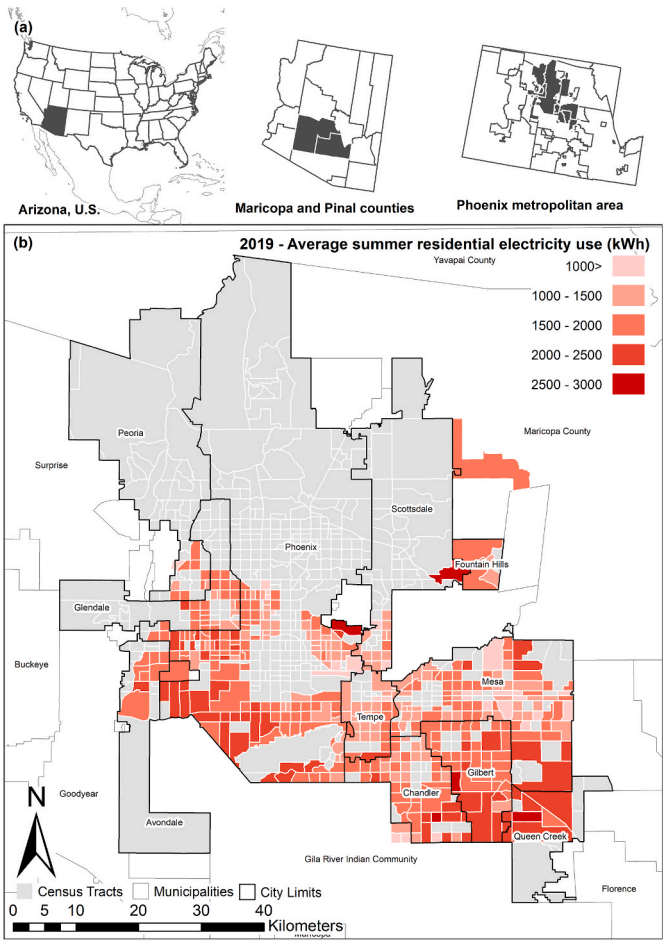


Fig. 1. Maps depicting (a) the location of the Phoenix Metropolitan Area (PMA) and (b) the spatial distribution of 2019 average summer residential electricity use (ASREU) at the census tract level.

3.2. Dependent and independent variables

The electricity consumption data used in this study were sourced from the Salt River Project (SRP), one of the two major utility companies servicing the PMA, alongside the Arizona Public Services Company (APS). These companies allocate electricity service based on census tracts; statistical subdivisions of a county designed to represent approximately between 2500 and 8000 people with similar socio-demographic characteristics [59].

SRP provided data for 426 census tracts in the PMA (Fig. 1b), covering nearly 48 % of the total tracts. Specifically, SRP provided the average residential household-level electricity consumption by census tract for the summer months of May, June, and July in 2019. Using 2019 data was driven by the availability and completeness of the SRP dataset. Additionally, it offers a pre-COVID baseline, unaffected by pandemic disruptions, which are not being considered in the objectives of this study. For this analysis, the average values of these three months was computed to yield the *2019 average summer residential electricity use* (ASREU) in kilowatt-hours (kWh), our designated response variable. The ASREU mean for the Phoenix region is 1614 kWh, with a standard deviation of 396.33. The range extends from a minimum of 361 kWh to a maximum of 3016 kWh, while the normal range, encompassing 95 % of residential electricity consumption, falls between 836.92 and 2390.49 kWh. This not only suggests a quasi-Gaussian distribution for ASREU, but also underscores significant variability in summer electricity consumption at the census tract level within the PMA.

The dataset encompasses 30 variables that capture a diverse array of demographic, social, and physical characteristics. Demographic and social information includes total population and density, population age distribution, employment and unemployment rates, education level percentages, household income, and racial composition (White, Black, Indian, Asian, Hispanic, Others). Additionally, the dataset incorporates physical descriptors of both urban surfaces and building structures. This includes specifics such as land cover fractions and building attributes like percentages of occupied house status, number of rooms, and building age. For this study, the 30 predictors were organized into five categories to improve the coherence and orderliness of subsequent analyses, especially during multicollinearity assessments. The categorization is as follows:

- **Demographic Variables:** Providing insights into population and socioeconomic variables.
- **Building Information Variables:** Describing basic physical features of buildings.
- **Racial Percentages:** Offering proportions of different racial categories.
- **Land Cover Percentages:** Representing the ratios of various land surface types.
- **Household Variables:** Furnishing information about the occupants of residential units.

Table 1 shows specific variables within each category along with their respective descriptive statistics and their main data sources.

4. Results and discussions

In the following section, we present the results based on the techniques described in the methods and modeling framework. First, we discuss the variance inflation factor analysis, highlighting high correlations and variable redundancies. Next, we introduce the initial standardized multivariate regression model, assessing its assumptions through outlier diagnostics with a focus on qualitative outlier characteristics. Then, we compare results from best subset selection and regularization techniques, examining differences in standardized coefficients and predictive performance. Lastly, we present a final multivariate model with centered predictors, interpreting coefficient impacts and discussing predictive performance, concluding with study limitations and future research opportunities.

4.1. Correlation and variance inflation factor analysis

To identify and address multicollinearity, we calculated Pearson correlation coefficients among variables, with a focus on identifying correlation values greater than 0.75 or lower than −0.75. Noteworthy, unsurprising correlations included those among land cover percentages. Specifically, Percent All Vegetation with percent Grass (0.83) and percent Trees (0.87). Correlations race percentages were also found in percent Hispanic and percent other races (0.77). None of the variables mentioned were removed from the dataset until a case of multicollinearity was confirmed.

Subsequently, a VIF analysis was conducted, revealing that no high VIF values were found within the demographic variables. Similarly, building information variables showed low VIF values, indicating independence. For racial percentages, percent white and percent other races had elevated VIF values, achieving VIF values of 315.31 and 146.75 respectively, leading to the removal of percent other races reducing dramatically VIF values. Land cover percentages exhibited multicollinearity, particularly between percent all vegetation (26.90) and percent trees (13.20). Eliminating percent all vegetation addressed this issue. For household variables, initially, no multicollinearity was detected, with household income having the highest VIF (6.36).

When fitting a multivariate linear model with all predictors (excluding previously dropped variables), new multicollinearity issues

Table 1
Overview and descriptive statistics for the dependent variable and the categories of variables employed in the analysis. Independent variables are grouped into demographic, building information, racial and land cover percentages, and household variables. Note that certain variables are subsequently eliminated during the correlation and variance inflation factor (VIF) analysis.

Variable	Min.	Median	Mean	Max.	SD.	Source
2019 average summer residential electricity use (ASREU) (kWh)	361	1657	1614	3016	396.33	SRP
Demographic variables						
Total Population (# persons)	1261	4823	5048	16890	1884.29	ACS
Population density (persons per square mile)	0.00	0.09	0.09	0.75	0.05	MAG
Percent with bachelor >25 years old (%)	1.05	28.69	30.14	75.42	17.41	ACS
Unemployment rate	0.00	4.53	5.14	15.40	2.88	ACS
Median age (years)	22.70	34.00	36.35	76.40	9.84	ACS
Building information variables						
Average house size (m^2)	25.20	154.70	169.80	1058.40	79.15	MAG
Average home age (years)	3.84	34.63	34.53	67.33	14.49	ACS
Median number of rooms	2.60	5.50	5.42	8.40	1.03	ACS
Racial percentages (%)						
Percent White	30.52	77.35	75.58	99.41	12.86	ACS
Percent Black American	0.00	4.62	6.44	29.79	5.98	ACS
Percent Indian native	0.00	1.12	2.11	36.12	2.95	ACS
Percent Asian	0.00	2.86	4.28	38.18	4.82	ACS
Percent two races	0.00	3.38	3.77	16.63	2.40	ACS
Percent Hispanic	1.13	25.00	34.11	92.87	23.75	ACS
Percent other races	0.00	4.39	7.54	52.33	8.64	ACS
Land cover percentages (%)						
Percent Buildings	0.89	17.48	16.89	32.19	5.88	CAP LTER
Percent Impervious	1.74	26.30	26.97	63.92	10.38	CAP LTER
Percent soil	6.68	31.19	32.73	69.36	11.03	CAP LTER
Percent grass	0.51	5.31	8.25	8.63	25.61	CAP LTER
Percent trees	0.87	9.33	9.73	30.57	5.18	CAP LTER
Percent all vegetation	2.71	18.30	18.97	43.80	8.10	CAP LTER
Percent pools	0.00	22.44	25.34	78.10	17.59	MAG
Percent water	0.02	0.48	1.02	10.74	1.59	CAP LTER
Household variables						
Household size (# persons)	1.54	2.91	2.93	5.04	0.66	ACS
Household income ($)	22398	62543	68321	166719	27917	ACS
Percent occupied HU (%)	56.88	92.52	90.72	100.00	6.83	ACS
Percent occupied HU with no vehicle (%)	0.000	3.86	6.16	32.78	6.52	ACS
Percent HU owner occupied (%)	3.03	62.77	60.04	97.15	21.21	ACS
HU owner occupied value ($)	10300	223650	244397	964200	118454	ACS
HU occupied rent ($)	630	1200	1263	3234	364.01	ACS

SD: Standard Deviation.
HU: Housing Units.
SRP: Salt River Project.
MAG: Maricopa Association of Governments (https://azmag.gov/).
ACS: US Census, American Community Survey (ACS) 2016–2020 5-year estimates (https://www.census.gov/programs-surveys/acs/data.html).
CAP LTER: Central Arizona–Phoenix Long-Term Ecological Research (https://sustainability-innovation.asu.edu/caplter/data/view/knb-lter-cap.685.1/).

emerged. Specifically, percent with BA exhibited a VIF of 11.12 due to its high correlation with four predictors, prompting its removal from the dataset. Furthermore, number of rooms and household income showed elevated VIF values (12.84 and 13.86, respectively). Despite their significance in explaining household electricity consumption, these variables contributed to multicollinearity. To avoid dropping these important predictors, percent occupied by the owner, rent cost, and house value were removed from the dataset. While this solution successfully eliminated multicollinearity, it led to number of rooms, percent Hispanic, household size and household income nearing the VIF threshold of 10. This strategic removal of variables aligns with recommended practices for addressing multicollinearity in statistical analyses as was suggested earlier. Table 2 presents the outcomes of the VIF analysis, indicating the effectiveness of the variable removal strategy in mitigating multicollinearity across different predictor categories.

This VIF analysis highlights the importance of addressing perfect correlation and redundancy among variables. While the percentage of different racial groups should sum to 100 % per census tract, the inclusion of Hispanic (or Latino) data introduces variability, as the American Community Survey (ACS) counts Hispanics within other racial categories, causing the total to exceed 100 %. This prevents perfect correlation and keeps VIF values from becoming infinite. Though multicollinearity was addressed by removing one race category, it's crucial to consider that some variables may measure similar attributes or directly sum to others, inflating VIF values. Overall, careful variable selection is essential to avoid redundancy and maintain the integrity of statistical analyses.

4.2. Elimination of outliers and checking ordinary least squares assumptions

After conducting the VIF analysis, the dataset retained a total of 24 predictors with no multicollinearity issues identified. A standardized multivariate linear model, incorporating all 24 predictors, was applied to detect and eliminate outliers while simultaneously assessing the assumptions of Ordinary Least Squares (OLS). In matrix form, this initial regression model can be expressed as:

$$\boldsymbol{Y}^* = \boldsymbol{X}_d^*\boldsymbol{\beta}_d^* + \boldsymbol{X}_b^*\boldsymbol{\beta}_b^* + \boldsymbol{X}_r^*\boldsymbol{\beta}_r^* + \boldsymbol{X}_l^*\boldsymbol{\beta}_l^* + \boldsymbol{X}_h^*\boldsymbol{\beta}_h^* + \boldsymbol{\epsilon}^* \tag{15}$$

In this equation, the column vector $\boldsymbol{Y}^*$ (n x 1), the standardized dependent variable, represents the 2019 Average Summer Residential Electricity Use (ASREU). The error term, $\boldsymbol{\epsilon}^*$ (n x 1), accounts for the random errors in the model. The matrices $\boldsymbol{X}^*$ (n x p) contain the 24 standardized predictor variables, which are grouped into five categories based on their nature. Specifically, $\boldsymbol{X}_d^*$ includes the four demographic variables, $\boldsymbol{X}_b^*$ represents the three building information variables, $\boldsymbol{X}_l^*$ comprises the six racial percentages, $\boldsymbol{X}_l^*$ corresponds to the seven land

Table 2
Summary of variance inflation factor (VIF) Analysis results for detecting multicollinearity: Within-category VIF values and overall dataset VIF values are presented. Grey shading indicates VIF values exceeding 10, while red shading highlights variables eliminated during analysis. Within-category VIFs represent the analysis among variables within each category, before and after eliminating some variables. Overall dataset VIFs encompass the analysis considering all predictors post within-category VIF analysis.

Variable	Within-category VIFs		Overall dataset VIFs.	
	Initial	Final	Initial	Final
Demographic variables				
Total Population	1.14	1.14	1.31	1.30
Population density	1.10	1.10	2.34	2.31
Percent with bachelor >25 years old	1.49	1.49	11.12	*Dropped*
Unemployment rate	1.33	1.33	1.49	1.47
Median age	1.26	1.26	5.68	4.24
Building information variables				
Average house size	1.18	1.18	1.75	1.74
Average home age	1.32	1.32	3.93	3.47
Median number of rooms	1.36	1.36	12.84	9.72
Racial percentages				
Percent White	315.31	5.42	5.95	5.75
Percent Black American	70.23	1.95	2.63	2.60
Percent Indian native	17.56	1.45	1.66	1.64
Percent Asian	45.18	1.95	2.68	2.50
Percent two races	12.30	1.27	1.40	1.40
Percent Hispanic	2.85	2.81	9.52	9.22
Percent other races	146.75	*Dropped*		
Land cover percentages				
Percent Buildings	1.73	1.67	4.46	4.38
Percent Impervious	3.64	3.62	6.01	5.41
Percent soil	3.51	3.37	3.91	3.82
Percent grass	7.72	1.66	1.79	1.76
Percent trees	13.20	2.31	2.88	2.82
Percent all vegetation	26.89	*Dropped*		
Percent pools	1.53	1.52	4.96	4.61
Percent water	1.09	1.09	1.22	1.19
Household variables				
Household size	1.57	1.57	9.90	8.08
Household income	6.35	6.35	13.86	7.66
Percent occupied HU	1.53	1.53	2.85	2.77
Percent occupied HU with no vehicle	2.04	2.04	2.50	2.18
Percent HU owner occupied	2.37	2.37	8.78	*Dropped*
HU owner occupied value	3.01	3.01	5.16	*Dropped*
HU occupied rent	3.17	3.17	3.55	*Dropped*

Table 3
This table presents a list of 25 outliers identified using studentized deleted residuals, leverage points, and Cook's distances. The observations included here are those that surpass absolute values of 2 for studentized deleted residuals, 0.2 for hat values, and 0.06 for Cook's distances. The outlier location can be seen in Fig. 2.

Outlier	Studentized residual	Hat value	Cook's distances
1	2.050	0.066	0.012
2	−1.316	0.762	0.222
3	−1.039	0.449	0.035
4	0.809	0.411	0.018
5	2.907	0.045	0.016
6	2.356	0.068	0.016
7	3.300	0.057	0.026
8	−3.445	0.080	0.040
9	−2.296	0.024	0.005
10	2.602	0.039	0.011
11	−2.565	0.066	0.018
12	2.373	0.069	0.016
13	2.356	0.035	0.008
14	2.859	0.094	0.033
15	−2.356	0.076	0.018
16	2.067	0.076	0.014
17	−2.048	0.348	0.089
18	7.738	0.063	0.141
19	3.762	0.130	0.082
20	5.058	0.051	0.053
21	−2.798	0.177	0.066

cover percentages, and $\boldsymbol{X}_h^*$ incorporates the four household variables. Likewise, the vectors $\boldsymbol{\beta}^*$ (p x 1) hold the standardized coefficients for the predictors, which are similarly grouped according to their respective categories, corresponding to the demographic (d), building information (b), racial percentages (r), land cover (l), and household (h) variables as outlined for the matrices $\boldsymbol{X}^*$.

Upon calculating metrics for outlierness, leverage, and influence for each observation in the dataset using the model expressed in equation (15), it was identified that 18 observations exhibited studentized deleted residuals greater than 2 or lower than −2. Additionally, 4 observations displayed leverage (Hat values) surpassing 0.2, and 5 observations had a Cook's distance exceeding 0.06.6 observations recorded high values across multiple metrics making a total of 21 identified regression outliers. A qualitative visual inspection using Google Maps was conducted on the identified outlying census tracts.

Fig. 2 shows the spatial location of these outliers and Table 3 the values of the corresponding metrics. Notably, the eighteen census tracts that displayed studentized deleted residuals with elevated absolute values, are mostly situated on the periphery of the urban area, characterized by residential condos, recent or ongoing development, and sizable parks or recreational spaces. Furthermore, four census tracts exhibited high Hat values, one including the Guadalupe area in south Tempe (outlier 4), where a significant portion of the population identifies with the Yaqui Mexican culture, the others census tracts featuring substantial green park with vegetation and lakes (outlier 2 and 17) and

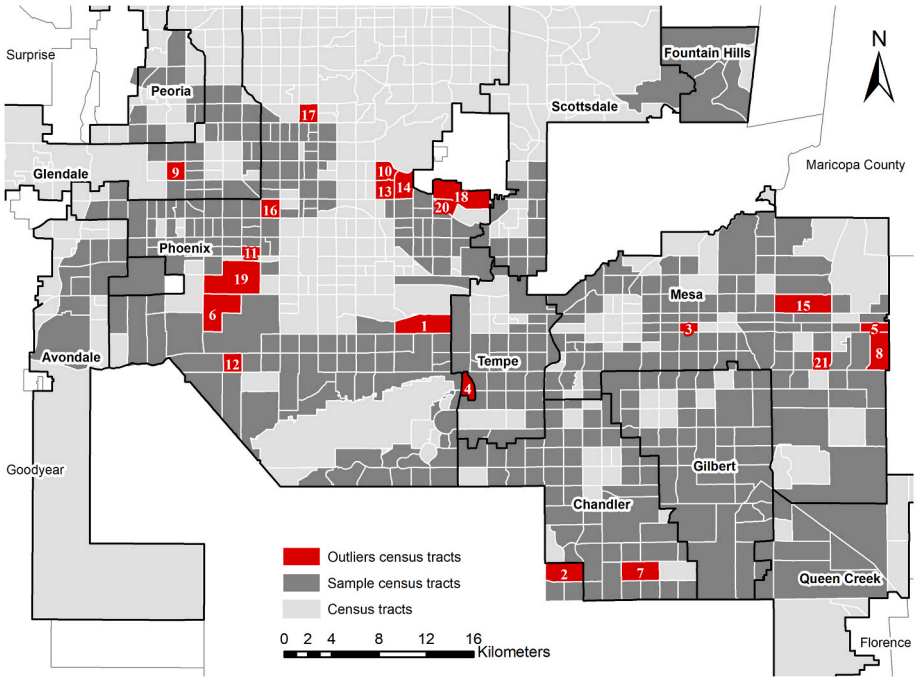


Fig. 2. The spatial location of regression outliers is indicated by numbering each census tract identified as an outlier. Table 3 displays the metrics and their respective values that demonstrate the degree of outlierness, leverage, and influence associated with each numbered outlier.

predominantly white-roofed residences (outlier 3).

All observations with high Cook's distance values also have elevated values in other metrics. Specifically, four census tracts showed both high Cook's distance and elevated absolute values of studentized deleted residuals. One of these census tracts is around the Comeback Mountain area in Phoenix (outlier 18), which is characterized by affluent neighborhoods, while the other is an industrial sector with minimal residential presence (outlier 19). Two census tracts showed high Cook's distance and Hat values, with one located in Phoenix's predominantly condominium zone (outlier 17) and the other situated on the outskirts of Chandler (outlier 2), also characterized by extensive green areas with vegetation and lakes as explained before. In summary, outliers are identified in census tracts on the urban periphery, characterized by industrialization or expansive green areas with lakes, uncommon in Phoenix. Cultural and economic factors might also contribute to these patterns. Removing these 21 census tracts restored normality and ensured consistent residuals. After eliminating outliers and refitting the model described in equation (15), the Shapiro-Wilk test confirmed that the residuals followed a normal distribution (p value = 0.31). Additionally, the Breusch-Pagan test, also known as the Non-constant Variance Score Test, indicated that the residuals exhibited homoscedasticity, with no signs of non-constant variance (p value = 0.82).

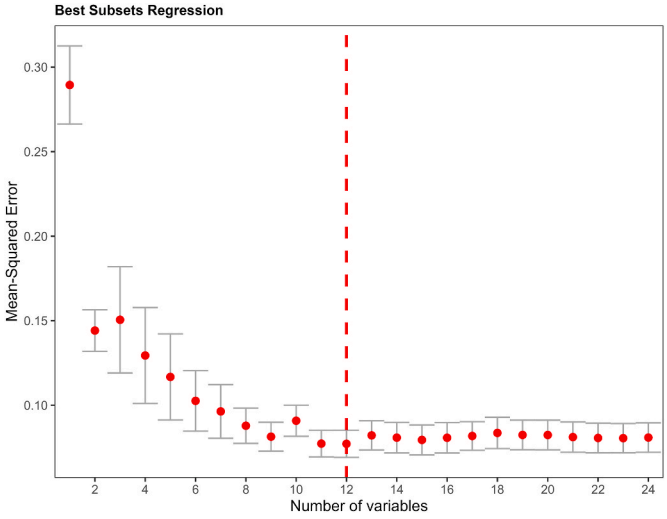


Fig. 3. Cross-validation LOOCV CVMSE plot of best subset selection method.

4.3. Variable selection and regularization procedures

To refine the selection process from the initial pool of 24 predictors, after multicollinearity analysis, we employed automatic selection and regularization procedures. These methods were used to identify the most relevant predictors and streamline the model, using the full model from equation (15) as the starting point. The outcome of the LOOCV cross-validation, depicted in Fig. 3, showcases the results of the automatic best subset selection procedure utilizing the branch-and-bound algorithm. While forward and backward selection was also executed, it yielded identical results and selected variables as best subset selection. The best subset procedure concluded that a model with 12 variables is superior, yielding a CVMSE of 0.077. It is noteworthy to note that the CVMSE progressively decreases until considering 9 variables. This suggests that including more than 9 variables in the model would not substantially improve its predictive performance.

Table 4 presents the standardized coefficients of the predictors for each selection procedure, alongside the coefficients from the regression model that includes the full set of predictors (full model – equation (15)) for comparison. Best subset technique identified variables such as the number of rooms, percentage of Hispanic population, home age, household size, percentage of pools, and household income as having the most substantial effects on ASREU (greater than 0.1), all but home age exhibiting a positive influence. Notably, there are minimal changes in coefficients between the best subset procedures and the coefficients of the full standardized regression model.

Fig. 4 illustrates the LOOCV cross-validation results for the Ridge, Lasso, and Elastic Net regularization methods. Traditionally, observing a

Table 4
Estimated Standardized coefficients of the selected variables obtained through best subset selection ($\widehat{\beta^*}_{Best}$), and Ridge ($\widehat{\beta^R}$), Lasso ($\widehat{\beta^L}$), and Elastic Net ($\widehat{\beta^E}$) regularization procedures. The first column displays coefficients from the full multivariate model expressed in equation (15) ($\widehat{\beta^*}_{Full}$).

Predictor	Estimated Standardized coefficients				
Demographic variables	$\widehat{\beta^*_{d\ Full}}$	$\widehat{\beta^*_{d\ Best}}$	$\widehat{\beta^R_d}$	$\widehat{\beta^L_d}$	$\widehat{\beta^E_d}$
Total Population	−0.059	−0.053	−0.028	−0.015	−0.018
Population density	0.014	–	0.027	–	–
Unemployment rate	−0.013	–	0.009	–	–
Median age	−0.034	–	−0.057	−0.009	−0.040
Building information variables	$\widehat{\beta^*_{b\ Full}}$	$\widehat{\beta^*_{b\ Best}}$	$\widehat{\beta^R_b}$	$\widehat{\beta^L_b}$	$\widehat{\beta^E_b}$
Average house size	0.090	0.096	0.131	0.113	0.127
Average home age	−0.102	−0.122	−0.039	−0.047	−0.035
Median number of rooms	0.436	0.439	0.265	0.393	0.293
Racial percentages	$\widehat{\beta^*_{r\ Full}}$	$\widehat{\beta^*_{r\ Best}}$	$\widehat{\beta^R_r}$	$\widehat{\beta^L_r}$	$\widehat{\beta^E_r}$
Percent White	0.017	–	−0.025	–	0.000
Percent Black American	0.025	–	0.002	0.001	0.004
Percent Indian native	−0.016	–	−0.037	–	−0.017
Percent Asian	−0.008	–	−0.014	–	–
Percent two races	0.000	–	−0.013	–	–
Percent Hispanic	0.316	0.313	0.155	0.182	0.162
Land cover percentages	$\widehat{\beta^*_{l\ Full}}$	$\widehat{\beta^*_{l\ Best}}$	$\widehat{\beta^R_l}$	$\widehat{\beta^L_l}$	$\widehat{\beta^E_l}$
Percent Buildings	−0.030	–	−0.023	−0.006	−0.002
Percent Impervious	−0.039	–	−0.092	−0.024	−0.082
Percent soil	0.061	0.087	0.072	0.076	0.074
Percent grass	−0.027	–	−0.022	–	−0.012
Percent trees	0.006	–	−0.023	–	−0.010
Percent pools	0.243	0.233	0.168	0.180	0.163
Percent water	−0.028	−0.024	−0.019	0.000	−0.009
Household variables	$\widehat{\beta^*_{h\ Full}}$	$\widehat{\beta^*_{h\ Best}}$	$\widehat{\beta^R_h}$	$\widehat{\beta^L_h}$	$\widehat{\beta^E_h}$
Household size	0.249	0.263	0.256	0.311	0.282
Household income	0.122	0.128	0.132	0.081	0.120
Percent occupied HU	0.067	0.082	0.119	0.089	0.113
Percent occupied HU with no vehicle	0.099	0.100	0.023	0.032	0.017

U-shape in plots of changes in the penalty term (λ) is indicative of the bias-variance tradeoff, as suggested by Hsu [7]. However, our results, as well as findings from other studies on electricity consumption modeling using regularization, do not exhibit this clear U-shape. This could be attributed to the fact that the optimal values of λ determined by the regularization techniques are relatively small. As Satre-Meloy [8] proposes, having several predictors that does not exceed the number of observations helps prevent overfitting, making an excessive amount of the penalty term unnecessary. In Fig. 4, two dotted lines represent values of λ. The first corresponds to the minimum, while the second is the λ within 1 standard error of the minimum, chosen as the optimal value. This choice is based in the pursuit of a more parsimonious model with fewer variables while also maintaining a minimal mean squared error as suggested by James et al. [38], and done by Satre-Meloy [12] and Hsu [7].

Table 4 illustrates the impact of Ridge regression on predictor coefficients compared to the full standardized multivariate model. Notably, the number of rooms and household size emerge as the most influential predictors, each with an effect greater than 0.2 on ASREU. It is intriguing to observe how the Ridge procedure diminishes the effect of the most crucial variable (number of rooms) from 0.44 to 0.27, while amplifying the effect of the second-highest variable (household size) from 0.09 to 0.13. This phenomenon may arise due to the remining correlation of the number of rooms with other predictors. Ridge regression effectively redistributes the regularization impact among them, highlighting the tendency of regularization procedures to penalize large coefficients. In essence, Ridge reduces coefficients by approximately 43 %. Unfortunately, Ridge does not lead to a parsimonious model, retaining all 24 predictors. As anticipated, Lasso and Elastic Net regression offer more parsimonious models. Lasso identifies the optimal model with 16 variables, while Elastic Net opts for one with 20 variables. The key distinction lies in Elastic Net incorporating additional racial and landcover variables, even though their impact on ASREU is less than 0.05. Consistently, important predictors identified by Lasso and Elastic Net, with effects greater than 0.1, include the number of rooms, household size, house size, percentage Hispanic and percentage of pools. Lasso's shrinkage effects reduce coefficients by around 35 %, while Elastic Net achieves a reduction of approximately 31 %.

To compare and choose a final model, we calculated in-sample and out-of-sample goodness-of-fit metrics for the models generated by each automatic selection and regularization method, as presented in Table 5. Notably, all models exhibit exceptional performance, achieving R^2 and adjusted R^2 values exceeding 0.90. Moreover, there's no indication of overfitting, as the differences in metrics between in-sample and out-of-sample are minimal (less than 0.03). Specifically, the highest in-sample R^2 is observed for the full standard regression model (0.94) and the model produced by the best subset algorithm (0.93). However, this is attributed to the high number of variables, when examining the adjusted R^2, their values are comparable to those of other procedures. Despite the findings of Hsu [7] and Satre-Meloy [8], in this analysis regularization methods do not necessarily outperform traditional regression methods. This discrepancy might be attributed to the dataset size, since regularization methods appear to excel in larger datasets with high variability [42] as several machine learning techniques [38]. Taking all these considerations into account, the selection of a final model leans towards parsimony. In this context, the regression model proposed by the best subset procedure stands out, providing the model with the fewest variables (12). Additionally, the traditional OLS multivariate regression model can be applied using the variables identified by best subset selection. This allows for inferences and testing on the coefficients, facilitating interpretation of the effects.

4.4. Residential electricity use trends and final model insights

A final multivariate regression model was developed, incorporating the 12 variables identified in the previous section. To preserve the original units of the predictors and improve the interpretability of their coefficients, all variables were centered around their means (see Equation (7)). The resulting model equation is:

$$\begin{aligned} ASREU_i = {} & \beta_0 + \beta_{h1}\left(\widetilde{X}_{h1,i}\right) + \beta_{h2}\left(\widetilde{X}_{h2,i}\right) + \beta_{h3}\left(\widetilde{X}_{h3,i}\right) + \beta_{h4}\left(\widetilde{X}_{h4,i}\right) + \beta_{b1}\left(\widetilde{X}_{b1,i}\right) \\ & + \beta_{b2}\left(\widetilde{X}_{b2,i}\right) + \beta_{b3}\left(\widetilde{X}_{b3,i}\right) + \beta_{l1}\left(\widetilde{X}_{l1,i}\right) + \beta_{l2}\left(\widetilde{X}_{l2,i}\right) + \beta_{l3}\left(\widetilde{X}_{l3,i}\right) \\ & + \beta_{d1}\left(\widetilde{X}_{d1,i}\right) + \beta_{r1}\left(\widetilde{X}_{r1,i}\right) + \varepsilon_i \end{aligned} \tag{16}$$

In this model, $ASREU_i$ represents the average summer residential electricity use for the i-th observation or census tract. $\widetilde{X}_{h1,i}$, $\widetilde{X}_{h2,i}$, $\widetilde{X}_{h3,i}$, $\widetilde{X}_{h4,i}$, represent the selected centered household variables (household size, household income, percent occupied housing units with no vehicle, and percent occupied housing units, respectively). Similarly, $\widetilde{X}_{b1,i}$, $\widetilde{X}_{b2,i}$, $\widetilde{X}_{b3,i}$, are the centered building information variables (median number of rooms, average home age, and average house size). The selected centered land cover percentage variables are given by $\widetilde{X}_{l1,i}$, $\widetilde{X}_{l2,i}$, $\widetilde{X}_{l3,i}$, which correspond to the percentages of pools, soil, and water. While $\widetilde{X}_{d1,i}$ corresponds to the remaining centered demographic variable (total population), and $\widetilde{X}_{r1,i}$ denotes the selected centered racial percentage variable (Hispanic). The intercept is represented by β_0, and the regression coefficients for the corresponding centered predictors are β_{h1}, β_{b1}, β_{l1}, β_{d1}, β_{r1}, and so on. Finally, ε_i represent the error term for the i-th observation. In this context, β_0 represents the estimated mean of ASREU

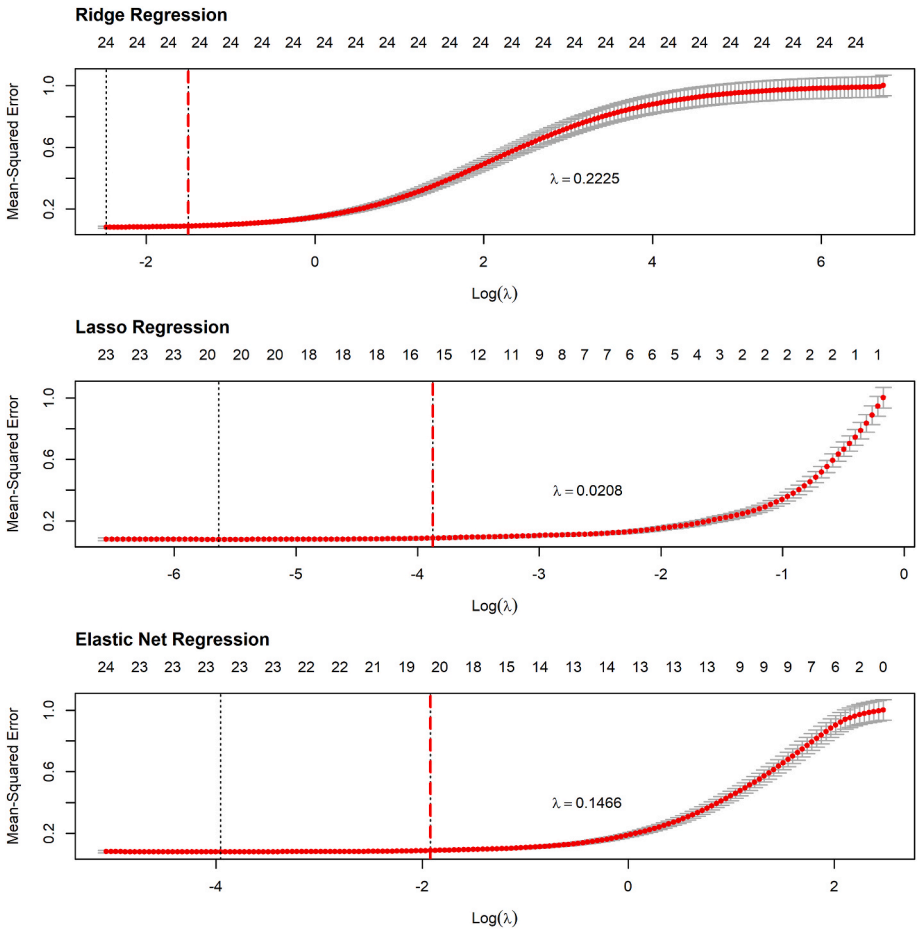


Fig. 4. Plots illustrating cross-validated prediction errors for Ridge, Lasso, and Elastic Net regression methods. The vertical axis shows CVMSE, while the bottom axis displays the logarithm of the tuning parameter (λ) for each method. The top axis indicates the number of non-zero coefficients in each model. Each curve's center line represents CVMSE from LOOCV, with grey shaded areas indicating the CVMSE standard error. Vertical dotted lines denote the optimum models selected based on minimum CVMSE (left) and the model with the fewest variables within 1 standard deviation of the minimum MSE (right, highlighted in red). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 5
In-sample and out-of-sample mean squared error (MSE), R-squared, adjusted R-squared, and the number of nonzero coefficients for the best models obtained from each variable selection and regularization methods. The full model refers to the standardized multiple regression described in equation (15).

Method	RMSE		R^2		Adjusted R^2		Nonzero Coefficients
	In sample	Out of sample	In sample	Out of sample	In sample	Out of sample	
Full Model	0.255	0.284	0.935	0.919	0.931	0.914	24
Ridge	0.281	0.298	0.921	0.915	0.916	0.909	24
Lasso	0.273	0.296	0.925	0.913	0.923	0.910	16
Elastic Net	0.279	0.312	0.922	0.911	0.918	0.906	20
Best Subset	0.259	0.277	0.933	0.923	0.931	0.920	12

in kilowatt-hours when all other variables are at their means. The coefficients quantify the change in ASREU for a one-standard-deviation shift in the predictor.

Additionally, separate regression models were developed for each category of variables; household, building information, demographic, racial, and land cover, allowing for a comparison of coefficient values, R^2, adjusted R^2, and partial $R^2_{YA|B}$. This approach provided a detailed assessment of the contribution of each variable category to the overall model, facilitating a more complete understanding of how different groups of predictors influence the model's performance and the dependent variable. These comparisons allowed for a comprehensive evaluation of the model's quality and robustness. These results are presented in Table 6.

Upon examination of R^2 and adjusted R^2 values, it is evident that household variables account for 84 % of the variation in ASREU, while building information variables explain 71 %, and land cover percentages elucidate 47 %. Conversely, when analyzed independently as predictors, the demographic variable of total population and the racial variable of percent Hispanics can only account for 5 % and 2 %, respectively. In terms of coefficients, focusing solely on selected building information variables reveals that the number of rooms and house size stands out as significant variables. However, this shifts when all 12 variables are incorporated into the model, with home age regaining significance. Notably, the sign of the effect for home age undergoes a change from initially positive to negative. A similar alteration is observed in the case of total population. This phenomenon can be interpreted as unmasking, wherein the inclusion of all 11 variables unveils a more accurate

Table 6
Coefficients of the final multivariate regression model along with associated goodness-of-fit metrics. To comprehensively assess the contribution of variables in explaining the variation in ASREU, sub-models containing only predictors within each category were also constructed. All predictors were centered to favor interpretability and conservate original units.

Predictor	$\widehat{\beta}$	Household	Building information	Land cover	Demographic	Racial	Full final model
Household size (0.657)	$\widehat{\beta}_{h1}$	350.493*** (13.311)					151.824*** (18.707)
Household income^{+} (2.808×10^4)	$\widehat{\beta}_{h2}$	78.876*** (3.473)					17.308*** (4.664)
Percent occupied HU with no vehicle (6.475)	$\widehat{\beta}_{h3}$	3.068** (1.494)					5.846*** (1.102)
Percent occupied HU (6.214)	$\widehat{\beta}_{h4}$	10.116*** (1.431)					5.022*** (1.087)
Median number of rooms (1.013)	$\widehat{\beta}_{b1}$		309.179*** (13.311)				164.515*** (12.521)
Average home age (14.469)	$\widehat{\beta}_{b2}$		1.374 (0.848)				−3.212*** (0.553)
Average house size (50.034)	$\widehat{\beta}_{b3}$		0.511* (0.414)				0.725*** (0.166)
Percent pools (17.493)	$\widehat{\beta}_{l1}$			10.907*** (0.797)			5.062*** (0.601)
Percent soil (10.965)	$\widehat{\beta}_{l2}$			14.734*** (1.258)			3.006*** (0.542)
Percent water (1.620)	$\widehat{\beta}_{l3}$			−0.197 (8.565)			−5.671* (3.163)
Total population (1868.133)	$\widehat{\beta}_{d1}$				0.048*** (0.010)		−0.011*** (0.003)
Percent Hispanic (23.664)	$\widehat{\beta}_{r1}$					2.369*** (0.790)	5.021*** (0.568)
Intercept	$\widehat{\beta}_0$	1,613.250*** (7.451)	1,613.250*** (10.098)	1,613.250*** (13.698)	1,613.250*** (18.341)	1,613.250*** (18.675)	1,613.250*** (4.958)
R^2		0.845	0.715	0.476	0.056	0.022	0.933
Adjusted R^2		0.844	0.713	0.472	0.054	0.019	0.931
Partial $R^2_{YA\|B}$		$R^2_{ASREU\ \ h}$ 0.845	$R^2_{ASREU\ h\|b}$ 0.353	$R^2_{ASREU\ h\|l}$ 0.267	$R^2_{ASREU\ h\|d}$ 0.025	$R^2_{ASREU\ h\|r}$ 0.007	–

Notes: Numbers in parentheses ( ) indicate the standard deviation for variables and the standard errors for coefficients. Note that these standard deviations differ from those in Table 1 due to the removal of extreme values and outliers in Section 4.2.
$R^2_{YA|B}$ is relative to the household variables model.
*p < 0.1; **p < 0.05; ***p < 0.01.
$^{+}$Household income was divided by 10,000, indicating that each unit increase in the coefficient of the regressor corresponds to a $1 $\times$ 10^4 increase in household income.

association between these variables and ASREU. This suggests that when only building information variables are considered, a highly correlated variable like the number of rooms may obscure the true effects of other predictors. Additionally, exclusive consideration of predictors with low predictive power, such as any demographic variable, may fail to uncover their true effects. Consequently, a diverse array of variables should be incorporated to adjust effect sizes and contribute to minimizing variation in the residuals.

A plausible reason is that older homes in PMA may rely more on natural gas for residential appliances like cooking and water heating, which reduces their electricity consumption. According to our model, an increase of 14.5 years in home age is associated with a decrease of approximately 3.2 kWh in residential electricity use, a marginal reduction possibly linked to these end uses. National data from the 2020 Residential Energy Consumption Survey (RECS) [60] supports this, indicating that about 20 % of households in Arizona use natural gas [61], with 40.4 % using it for water heating and 22.9 % for cooking [62]. While electrification efforts are ongoing, natural gas remains common in older homes [63]. Future research should investigate these possible trends further specifically for PMA, including their implications for carbon emissions and the uneven progress of home electrification. Another potential explanation for the negative effect of home age could be related to energy efficiency retrofits in older homes.

The negative relationship between total population and ASREU suggests that as population density increases within a census tract, individual household electricity consumption tends to decrease. This could be attributed to a higher proportion of multi-family dwellings (e.g., apartments and condos) in densely populated areas, which generally have lower per-unit electricity consumption compared to single-family homes [64]. Multi-family units often share walls and roofs, leading to more efficient energy use for heating, cooling, and lighting [65]. Therefore, the model's result suggesting that an increase of 1868 inhabitants in population decreases ASREU by 0.011 kWh aligns with these patterns of urban energy use efficiency.

To quantify the relative contribution of the predictive power for the different categories of variables, the partial $R^2_{YA|B}$ was calculated relative to the R^2 obtained only using the household variables (0.84). For the building information variables, a partial $R^2_{ASREU\ h|b}$ of 0.35 was obtained. This can be interpreted as the SSE of a model encapsulating only the selected household variables, can be reduced by 35 % if the selected building information variables are added. Similar interpretation can be done with the land cover percentage, if percent of pools and percent of soil are added, the amount of variation that the model can explain can increase (reduce SSE) by 27 %. The inclusion of total population and percent Hispanic reduce SSE only by 3 % and 0.7 % respectively. Confirming the great predictive performance that household and building information variables have on ASREU. However, this does not mean that demographic or racial variables should not be included. As previously discussed, hidden patterns can emerge, as seen in the case of the percentage of Hispanic population, which consistently reflects a specific electricity consumption trend among Hispanic communities in Phoenix.

This assumption is supported by the consistent relevance of this variable in all preceding procedures outlined in the previous section.

In essence, the parameters coefficients of the full final model serve as a powerful tool for elucidating residential electricity consumption dynamics during the summer. This model accounts for 93 % of the variation in ASREU, and of paramount importance, its coefficients are interpretable. Analyzing these coefficients, the centered predictors unveil significant associations. Holding other variables constant, a one-standard-deviation increase of 0.66 persons in household size correlates with a noteworthy average increase of approximately 151.8 kWh in summer electricity use. Likewise, an augmentation of $28,080 in annual household income translates to an increase of 17.3 kWh in ASREU. Notably, the addition of just one room dramatically amplifies electricity consumption on average by 164.5 kWh during the summer. Furthermore, a 24 % increment in the Hispanic population within a census tract is linked to a notable increase of 5 kWh in ASREU. These kinds of interpretations can be obtained by the findings in Table 6.

It is essential to note that ASREU represents a summer monthly average, meaning the coefficient interpretations in Table 6 reflect a typical seasonal variation in ASREU during the summer months. The model is particularly useful for capturing the intricate relationships between sociodemographic factors and residential electricity consumption. By doing so, it provides valuable insights into patterns of energy use and allows for reasonably accurate predictions of household electricity consumption within the PMA. However, it is important to clarify that the model is not designed to predict absolute electricity consumption levels at the census tract level. Since ASREU represents an average for households, any predictions made using this model should be interpreted as the typical household consumption within each census tract, not the total or absolute electricity use for the entire tract. This distinction is crucial for understanding the scope and applicability of the model's predictions. The model's precision in quantifying the average impact of various socioeconomic factors aids energy policy makers and practitioners in making informed decisions. It helps identify target socioeconomic groups and determine effective control variables when evaluating specific policies in the residential sector. Accurate policy assessment is crucial for achieving urban sustainability goals, and these results offer clear insights into influential variables and their role in modeling residential electricity use.

4.5. Limitations and research opportunities

This project's limitations in design, data, and methods merit consideration for future studies. The incomplete representation of housing characteristics and lack of appliance ownership details, influential for electricity consumption in general [66], suggest avenues for improvement. Also, recent research recalls the need to start analyzing future changes in residential building stock driven by individual and household behaviors [67]. Future studies could explore these specifics to reveal targeted patterns for cooling energy demand reduction.

While the primary aim of this study was to explore the relationship between sociodemographic factors, some building features, land cover contextual factors and residential electricity consumption at the census tract level, it is crucial to recognize the role of climate variability in shaping electricity use. Specifically, understanding how local climate factors interact with building cooling demand is essential for examining the impact of urban heat island (UHI) intensities on spatial variations in summer electricity consumption. Currently, the absence of detailed climate variables among census tracts limits a more comprehensive understanding of these effects. Including such variables would not only deepen insights into climate related electricity consumption but also clarify how intra-state climate variations affect state-level electricity trends, as has been explored by Sailor et al. [68] or Staffell et al. [69], and neighborhood-scale disparities, as shown by Salvati et al. [70] and Luo et al. [71]. However, given the cross-sectional nature of the dataset used in this study, any climate data incorporated would need to be aggregated both spatially and temporally, significantly reducing the variability across census tracts. Moreover, climate variables tend to exhibit high collinearity, which suggests that a broader set of variables, such as humidity and wind speed, must be considered to mitigate omitted variable bias [72]. Future research should integrate a panel data approach, combining climate predictors with key sociodemographic variables identified in this study, to more accurately assess the influence of local climate variations on residential electricity use in an urban area.

In this study, variables were aggregated to the census tract level, a common practice known to introduce spatial autocorrelation. The Global Moran's I test was used to identify spatial autocorrelation in ASREU, revealing a Moran's Index of 0.25 with a z-score of 29.96, indicating spatial clustering. However, analyzing the residuals of the final model showed a Moran's Index of 0.007 and a z-score of 1.12, suggesting spatially random distribution, aligned with expectations given the model's explanatory power. Thus, methods like Geographically Weighted Regression were unnecessary. While this study did not require spatial analysis, future research could add insight by exploring spatially varying relationships among crucial ASREU predictors.

The replicability of this study hinges on clear methodological transparency and consistent data collection practices, which would provide new insights due to environmental conditions, housing stock, sociodemographic, and energy policies in the different regions. The replicability of this study in other cities and countries is highly feasible due to the widespread availability of sociodemographic and energy consumption data, though specific variables influencing residential electricity use may vary by region. Many countries conduct regular surveys or censuses like the U.S. American Community Survey (ACS), which provide detailed household-level data. For instance, the European Union's Household Budget Surveys and energy data from Eurostat [73] offer comparable datasets for modeling energy consumption patterns in European nations. Additionally, the International Energy Agency (IEA) provides data across 24 countries, including most of Europe, the U.S., and Australia [74], facilitating cross-country analysis of residential energy use by end use. A potential challenge could be obtaining local utility electricity data, as was done in this study, but with proper anonymization protocols and industry collaboration, other researchers can acquire similar datasets. Therefore, re-estimating the model's parameters in different local contexts is not only possible but encouraged to foster comparison and further validate the techniques used in other contexts.

5. Conclusions

This study applied regression methods to analyze how sociodemographic and land cover variables influence residential electricity usage per household (ASREU) in Phoenix, Arizona. We addressed multicollinearity and outliers, then used automatic variable selection and regularization techniques with 24 potential predictors to develop a concise model. Through cross-validation, we assessed the final model's goodness of fit, identifying influential predictor categories and interpreting coefficients.

Exploring multicollinearity in this study provides light on the complexities of inter-variable relationships. It underscores the limitations of relying solely on correlation analysis and emphasizes the importance of additional metrics like the Variance Inflation Factor (VIF) for a thorough assessment. The study highlights how variables presented as percentages, such as land cover ratios and racial percentages, can contribute to multicollinearity and emphasizes the need to recognize this vulnerability to avoid redundancy in regression models. Moreover, it reveals instances where multicollinearity extends beyond variable categories, particularly with high-variability predictors like household income.

The analysis unveiled outliers mainly located in census tracts bordering the urban periphery, marked by distinct land features like industrial zones or expansive green areas with lakes, which are unusual in the PMA. Cultural and economic factors also played a role in outlier

identification. Following outlier removal, statistical tests validated the efficacy of these procedures, indicating normal distribution and consistent residual variance. Additionally, visual inspection using satellite images added valuable context to the quantitative findings, highlighting the importance of qualitative procedures during outlier identification. Our analysis of variable selection and regularization procedures highlights the significant impact of household and building information variables on ASREU. Key variables such as number of rooms, household size, house size, percent Hispanic, and percent pools exhibit the most substantial standardized effects on ASREU. While regularization methods are effective, particularly in larger datasets with higher variability, our study emphasizes that these methods may not outperform traditional regression techniques in relatively small datasets with linear relationships and modest complexity, excluding quadratic and interaction terms. The selection of the final model, guided by in-sample and out-of-sample goodness-of-fit metrics, prioritizes parsimony, resulting in a regression model with 12 variables derived from best subset selection.

In the final regression model for ASREU during the summer of 2019, centered predictor variables reveal insightful associations with electricity consumption. Higher median household income, percent of pools, percent of soil, number of rooms, percentage of occupied housing units with no vehicle, and house size are linked to increased electricity use, suggesting that households with greater financial means and specific housing characteristics tend to exhibit higher electricity consumption. Conversely, the negative effect of home age on ASREU may stem from smaller home sizes with fewer rooms, energy efficiency retrofits in older homes and the use of natural gas for appliances like water heating and cooking, reducing electricity demand. Similarly, the negative relationship between total population and ASREU can be explained by higher-density areas with multi-family dwellings, which are more energy-efficient due to shared walls and roofs, leading to lower electricity use per household. Household and building information variables can explain over 70 % of the variation in ASREU, while demographic and racial variables contribute less to reducing unexplained ASREU variation. Nonetheless, these variables play a crucial role in revealing underlying patterns, adjusting coefficients, and controlling residual variability.

The interpretable regression coefficients of the final model illustrate the relationships between centered predictor variables and residential electricity use. The intercept term represents the estimated mean electricity use when all predictors are at their centered means, calculated at 1,613.250 kW-hours per household. Each coefficient indicates the change in electricity use associated with a one-standard-deviation change in the respective predictor, offering a quantitative insight into the impact of various household, building, land cover, and demographic factors on electricity consumption. This comprehensive model enhances understanding of the complex relationships of residential electricity use and facilitates precise predictions that could aid policy evaluation. This is essential for designing and maintaining, and improving public policies related to the residential sector. The results presented here convey a clear message about which variables have the greatest effect and should also be considered as controls when modeling residential electricity use.

This study pinpoints key areas for future research on summer residential electricity consumption, including improving housing representation, incorporating appliance ownership details, and accounting for local climate variations. Exploring these variables can refine cooling energy demand reduction strategies. Investigating interactions and quadratic terms in predictive models may unveil intricate patterns among predictors. Further research on spatially varying relationships among crucial predictors is warranted despite effectively capturing spatial variation. Ongoing exploration of spatial patterns could offer additional insights for informed decision-making in energy policy.

CRediT authorship contribution statement

Edwin Alejandro Ramírez-Aguilar: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **David J. Sailor:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Elizabeth A. Wentz:** Writing – review & editing, Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Edwin Alejandro Ramirez-Aguilar and David J. Sailor reports financial support was provided by US Department of Energy. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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